

Evaluating Retrieval-Augmented Verification for Scientific Claims

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Abstract

Automated fact verification systems increasingly rely on retrieval-augmented pipelines to ground claims in evidence. In this work we present a hybrid verification architecture combining sparse and dense retrieval with an ensemble of natural language inference models. We show that citation-graph reasoning further improves precision on a benchmark of scientific claims.

1. Introduction

Citation misuse is a persistent problem in scientific writing [1]. Prior studies have shown that a significant fraction of citations do not support the claims they are attached to [2, 3]. This finding has been replicated across multiple domains (Smith et al., 2020).

Automated citation verification has therefore attracted growing interest. Early approaches relied on simple lexical overlap between a claim and its cited source [4]. More recent work uses neural entailment models to assess whether a source actually supports a claim [5, 6].

2. Related Work

Evidence retrieval for fact verification has been studied extensively in the context of the FEVER shared task [7]. Cross-encoder re-ranking has been shown to substantially improve retrieval precision over bag-of-words baselines [8].

Natural language inference models, particularly those fine-tuned on MNLI, have become the standard tool for entailment-based verification [9, 10].

3. Methodology

Our pipeline consists of four stages: document processing, evidence retrieval, verification, and reporting. Document processing extracts claim-citation pairs from the source PDF, preserving page and paragraph provenance for later auditing.

Evidence retrieval combines BM25 with a dense SBERT retriever, followed by cross-encoder re-ranking of the top candidates. Verification applies an ensemble of two MNLI models and a numerical consistency checker for claims involving statistics.

4. Results

On our held-out evaluation set, the hybrid pipeline achieved 87.3% agreement with human annotators, compared to 74.1% for a lexical-overlap baseline. Citation-graph reasoning improved precision by an additional 4.2 points on claims with multiple grouped citations.

5. Discussion

These results suggest that combining retrieval-based evidence gathering with graph-based cross-citation checks meaningfully improves verification quality, particularly for claims supported by grouped citations rather than a single source.

6. Conclusion

We presented a hybrid citation verification pipeline and demonstrated its effectiveness on a benchmark of scientific claims. Future work will explore extending the approach to multi-document claim verification.

References

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